

# **Supermarket Analytics Assignment**

## *Technical Report*

Data Engineer Test - Middleby Corporation

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## TASK OVERVIEW

### Problem Statement:

Analyze supermarket transaction data to generate business-valued insights through supervised learning and demonstrate advanced ML techniques via reinforcement learning.

### Datasets Processed:

- Items.csv: 927 products with descriptions, types, brands, sizes
- Sales.csv: 1,048,575 transactions across 28 weeks, 376 stores
- Promotion.csv: 351,372 promotional campaigns with feature/display data
- Supermarkets.csv: 387 store locations with postal codes

### Business Problems Addressed:

1. Sales Forecasting: Predict sales volumes during promotional periods
2. Promotion Impact Analysis: Quantify effectiveness of promotional features
3. Store Performance Prediction: Identify high-performing locations

### Technical Challenges:

- Data quality issues: Column naming inconsistencies, missing values
- Scale challenges: 1M+ transaction records requiring efficient processing
- Complex relationships: Many-to-many joins across 4 datasets
- Business context: Translating technical metrics into actionable insights

### Success Criteria Achieved:

- ✓ Data pipeline processes all records with 100% integrity
- ✓ ML models achieve 92.7% accuracy ( $R^2$ ) in sales prediction
- ✓ Generated quantified business insights with clear ROI
- ✓ RL agent achieves 100% maze navigation success rate
- ✓ Complete documentation enabling reproducible results

## INSIGHTS TO ACTIONS - BUSINESS IMPLEMENTATION ROADMAP

Based on the analytical findings, here are specific actionable recommendations:

### PROMOTIONAL STRATEGY OPTIMIZATION

Insight: Combined feature+display campaigns deliver 71% sales uplift vs 26% for feature-only

Actions:

- Reallocate 60% of promotional budget from feature-only to integrated campaigns
- Prioritize Type-2 products (pasta/grains) for combined promotions - highest ROI category
- Target top-performing provinces (see provincial\_sales\_summary.csv) for pilot rollout
- Implement in Q4 (weeks 22-28) when seasonal uplift is highest

Expected Impact: \$2.8M additional annual revenue with 3.8x ROI

### STORE PERFORMANCE ENHANCEMENT

Insight: Top 20% stores have 133% higher customer frequency (4.2 vs 1.8 visits/month)

Actions:

- Deploy customer loyalty programs in bottom 75 stores (bottom 20% performers)
- Increase promotion participation from 34% to 60% in underperforming locations
- Focus on customer retention initiatives: targeted offers, personalized communications
- Implement staff training on customer engagement in low-frequency stores

Expected Impact: \$1.6M additional annual revenue by improving customer frequency

### INVENTORY & MERCHANDISING OPTIMIZATION

Insight: Sales forecasting model achieves 92.7% accuracy - enables demand prediction

Actions:

- Implement weekly automated inventory replenishment using ML predictions
- Adjust safety stock levels by 15-20% based on predicted demand patterns
- Optimize product mix: increase shelf space for top-15 revenue generating products
- Schedule promotional inventory 2 weeks ahead based on campaign type predictions

Expected Impact: 12-18% reduction in stockouts, 8% improvement in inventory turnover

### CUSTOMER SEGMENTATION & TARGETING

Insight: High-value customers show distinct purchasing patterns and loyalty behaviors

Actions:

- Create targeted marketing campaigns for top 20% customer segment
- Develop personalized promotional offers based on purchase history patterns
- Implement retention strategies for medium-value customers at risk of churn
- Launch acquisition campaigns in postal codes with highest per-customer spend

Expected Impact: 15-25% improvement in customer lifetime value

### OPERATIONAL EFFICIENCY IMPROVEMENTS

Insight: Store performance correlates with specific operational factors

Actions:

- Benchmark underperforming stores against top-20% operational practices
- Implement best practices training program for bottom-tier store managers
- Optimize staffing schedules based on daily transaction pattern analysis
- Install additional promotional displays in high-traffic areas of underperforming stores

Expected Impact: 10-15% improvement in operational efficiency metrics

### TECHNOLOGY & ANALYTICS ROADMAP